Log Book Entry 6.1

NB2420 Electronic Instrumentation · Manual 6.1 · Relaxation oscillator

Hand in via BrightSpace

Name(s): Partner:

Date: Group:

PREPARATION

complete before the practicum

1. *What theories, laws, and/or formulas are most important for this experiment?*
2. *What do you expect to learn?*
3. *What do you expect to observe, and why?*

COMPARATOR AND INTEGRATOR

Tasks B1, B2, A1, A2

Assume $U_H = +10V$ and $U_L = -10V$ for your first predictions. You measure the actual output limits later.

Integrator slopes Task B2, with $R_c = 47\text{k}\Omega$ and $C = 100\text{nF}$

Slope for an input of $+2V$:

Slope for an input of $-2V$:

Switching thresholds Task A2, with $R_a = 10\text{k}\Omega$ and $R_b = 33\text{k}\Omega$

Upper threshold $U_{\text{T,up}}$:

Lower threshold $U_{\text{T,down}}$:

Measured output limits U_H and U_L:

Why hysteresis? *In one or two sentences: what does the second threshold buy you when the input is noisy and slow?*

ONE OSCILLATION CYCLE

Tasks B3, A3, S1, S2

Predicted cycle Task A3

Slope of U_{tri} while $U_q = U_H$:

Slope of U_{tri} while $U_q = U_L$:

Duration of one ramp:

Period T and frequency f:

Simulated period and frequency:

What changes the cycle? Task A3 step 4

Change	Ramp slope	Triangle peak-to-peak	Period T
Double R_c			
Halve R_a			

CONCLUSIONS

Tasks B3 and A3

The loop in your own words *Start from a positive comparator output and follow one full cycle. Name the threshold that ends each ramp, and say where the energy comes from.*

Graph *Sketch U_q and U_{tri} against time for one complete cycle. Mark both switching thresholds and the period T.*

REFLECTION

complete before submission

1. *Which results matched or differed from your expectations, and why?*
2. *What is the most important thing you learned?*
3. *What would you do differently next time?*

TA check-off: Date: